Agentic AI with LangChain and LangGraph

Presentation 8: Intro to Deep Agent and Miscellaneous Items
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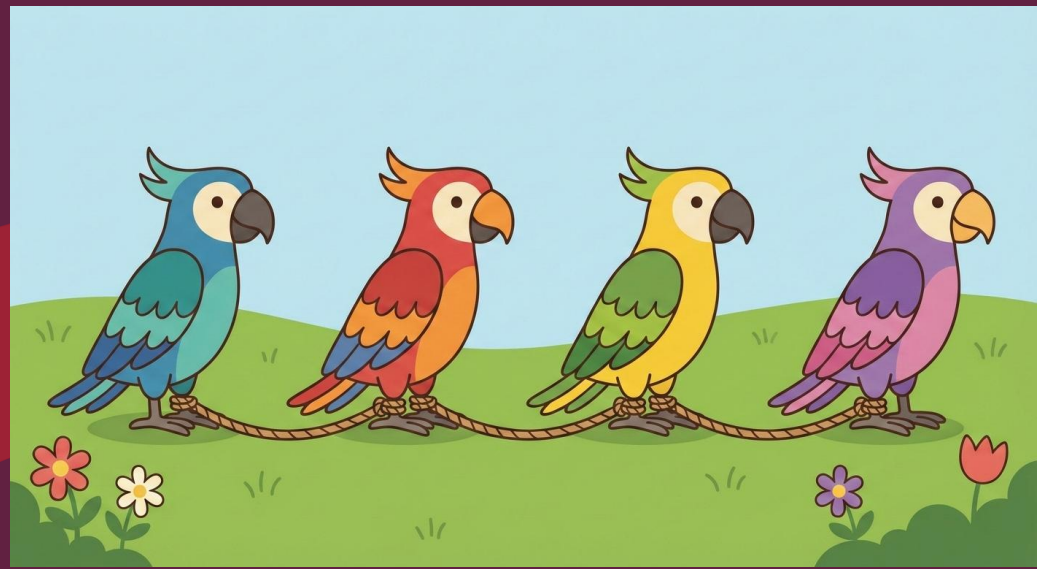

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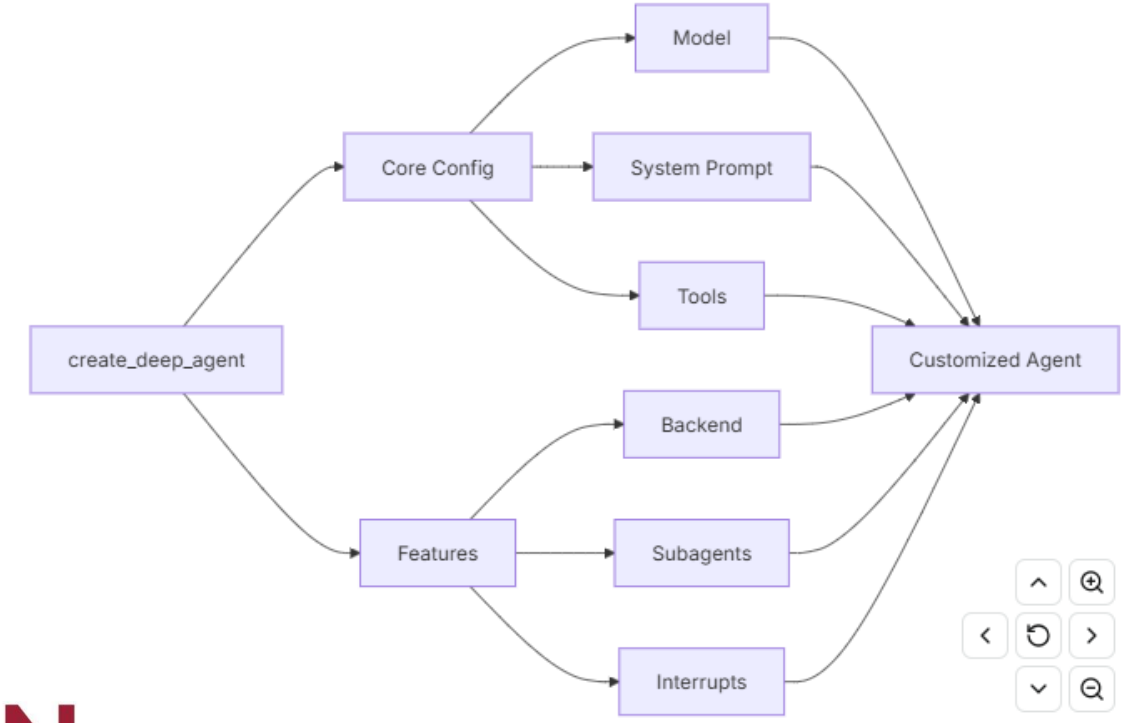
0. Deep Agent

0. Deep Agent: Quickstart (Skipped)

Deep Agent = regular ReAct agent + a list of langchain-provided default tools + planning capabilities + context management through file system……

According to LangChain docs, it seems Deep Agent is ReAct Agent Pro Max.

Since Deep Agent is really similar to regular ReAct agent, I won't repeat what I discussed before but talking something new.



0. Deep Agent: Subagents

You could provide some subagents to main Deep Agent so that main Deep Agent could focus on high-level planning and task decomposition and subagents could be called to work on specific sub-tasks.

You could add a dictionary, which is basically agent configuration file in the list and main Deep Agent would create the agent behind the scene, or a complied LangGraph graph (the StateGraph we discussed in the “Intro to LangGraph”, such as agentic workflow or multi-agents).

(Unfortunately, the example provided by LangChain is not that good but you could get it.)

I would go over the docs for you very quick.

1. Agent Skills

1. Agent Skills: What is it?

It's mainly advertised by Anthropic.

TL;DR: agent calls the tool to load information from files it needs to respond to user query. The information is basically skill.

It looks like RAG but using tool to retrieve text information it needs, rather than embedding model and vector store.

I would go over the example for you very quick. https://docs.langchain.com/oss/python/langchain/multi-agent/skills-sql-assistant

OpenAI
https://developers.openai.com › codex › skills

Agent Skills

Use **agent skills** to extend Codex with task-specific capabilities. A skill packages instructions, resources, and optional scripts so Codex can follow a ... Read more

Agent skill definition | Where to save skills | Create a skill

agentskills.io
https://agentskills.io › ...

Agent Skills: Overview

Agent Skills are folders of instructions, scripts, and resources that agents can discover and use to do things more accurately and efficiently. . Why Agent ... Read more

Visual Studio Code
https://code.visualstudio.com › copilot › customization

Use Agent Skills in VS Code

Dec 17, 2025 — **Agent Skills** are folders of instructions, scripts, and resources that GitHub Copilot can load when relevant to perform specialized tasks. Agent ... Read more

LangChain Blog
https://blog.langchain.com › using-skills-with-deep-age...

Using skills with Deep Agents

Nov 25, 2025 — **Skills** enable powerful capabilities beyond reducing token usage. As Barry Zhang of Anthropic mentions, **skills** are a step toward continuous ... Read more

Claude
https://code.claude.com › docs › skills

Agent Skills - Claude Code Docs

This guide shows you how to create, use, and manage **Agent Skills** in Claude Code. For background on how Skills work across Claude products, ... Read more

2. Research Frontiers in AI Agent

2. Research Frontiers in AI Agent: Multi-Modal Agent

The agent we have learned so far could only handle plain text……

Why is Multi-Modal Agent hard to build?
(I might ask a lot of questions….)

xAI

Back to jobs

New

Member of Technical Staff, Image Generation - Agent, RL

Apply

Palo Alto, CA; Seattle, WA

About xAI

xAI's mission is to create AI systems that can accurately understand the universe and aid humanity in its pursuit of knowledge. Our team is small, highly motivated, and focused on engineering excellence. This organization is for individuals who appreciate challenging themselves and thrive on curiosity. We operate with a flat organizational structure. All employees are expected to be hands-on and to contribute directly to the company's mission. Leadership is given to those who show initiative and consistently deliver excellence. Work ethic and strong prioritization skills are important. All engineers are expected to have strong communication skills. They should be able to concisely and accurately share knowledge with their teammates.

About the Role

The omni team at xAI creates magical AI experiences beyond text, enabling understanding and generation of content across various modalities, including image, video, and audio.

We are building next-generation multimodal agents that reason, act, and communicate through images as naturally as text. In many real-world scenarios, images are a more intuitive and efficient way to convey information than language alone. This role focuses on advancing image generation models with agentic behavior and reinforcement learning, enabling higher factuality, controllability, and seamless integration into multimodal conversational systems.

You will work on core research and production systems that allow models to decide when and how to generate images, improving Grok multimodal responses where images meaningfully augment or replace text.

2. Research Frontiers in AI Agent: Long Term Memory

Long Term memory is still not mature…

What do you think about how long-term memory is supposed to work in AI agent?

Amazon Web Services (AWS)
https://aws.amazon.com › blogs › machine-learning › b...

Building smarter AI agents: AgentCore long-term memory ...

Oct 15, 2025 — In brief, AgentCore **Memory** is a fully managed service that enables developers to build context-aware AI **agents** by providing both short-**term** ... Read more

Docs by LangChain
https://docs.langchain.com › oss › python › deepagents

Long-term memory - Docs by LangChain

You can extend deep **agents** with **long-term memory** by using a CompositeBackend that routes specific paths to persistent storage. This enables hybrid storage ... Read more

Microsoft Learn
https://learn.microsoft.com › en-us › user-guide › agents

Agent Chat History and Memory

Nov 10, 2025 — **Long term memory** ... The **Agent** Framework allows developers to provide custom components that can extract memories or provide memories to an **agent**. Read more

Chat History | Long term memory

3. Career Opportunities in LangChain Ecosystem

3. Career Opportunities in LangChain Ecosystem: These Companies…..

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Case studies

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This list of companies using LangGraph and their success stories is compiled from public sources. If your company uses LangGraph, we'd love for you to share your story and add it to the list. You're also welcome to contribute updates based on publicly available information from other companies, such as blog posts or press releases.

Company	Industry	Use case	Reference
AirTop	Software & Technology (GenAI Native)	Browser automation for AI agents	Case study, 2024
AppFolio	Real Estate	Copilot for domain-specific task	Case study, 2024
Athena Intelligence	Software & Technology (GenAI Native)	Research & summarization	Case study, 2024
BlackRock	Financial Services	Copilot for domain-specific task	Interrupt talk, 2025
Captide	Software & Technology (GenAI Native)	Data extraction	Case study, 2025
Cisco CX	Software & Technology	Customer support	Interrupt Talk, 2025
Cisco Outshift	Software & Technology	DevOps	Video story, 2025; Case study, 2025; Blog post, 2025
J.P. Morgan	Financial Services	Copilot for domain-specific task	Interrupt talk, 2025
Klarna	Fintech	Copilot for domain-specific task	Case study, 2025
Komodo Health	Healthcare	Copilot for domain-specific task	Blog post
LinkedIn	Social Media	Code generation; Search & discovery	Interrupt talk, 2025; Blog post, 2025; Blog post, 2024
Minimal	E-commerce	Customer support	Case study, 2025
Modern Treasury	Fintech	GenAI embedded product experiences	Video story, 2025
Monday	Software & Technology	GenAI embedded product experiences	Interrupt talk, 2025
Morningstar	Financial Services	Research & summarization	Video story, 2025
OpenRecovery	Healthcare	Copilot for domain-specific task	Case study, 2024
Pigment	Fintech	GenAI embedded product experiences	Video story, 2025
Prosper	Fintech	Customer support	Video story, 2025
Qodo	Software & Technology (GenAI Native)	Code generation	Blog post, 2025

4. Quick Review

(You can ask me some questions about what I covered so far)

Thank you for your paged attention!

arXiv > cs > arXiv:2309.06180

Computer Science > Machine Learning

[Submitted on 12 Sep 2023]

Efficient Memory Management for Large Language Model Serving with PagedAttention

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, Ion Stoica